Lecture 8: Integration of Rational Functions by Partial Fractions

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Apply the long division to an improper rational function.
- ☐ Find the partial fraction decomposition of a proper rational function.
- ☐ Integrating rational functions by partial fractions.

Introduction

A rational function is a function of the form $\frac{f(x)}{g(x)}$, where $f(x)$ and $g(x)$ are two polynomials, with $g(x) \neq 0$.

Eg: $\frac{2x}{x^2+4}$ ← deg = 1, ← deg 2

Eg: $\frac{x^2+4}{2x}$ ← deg = 2, ← deg = 1

$2x \overline{)\, x^2+4}$ quotient $\frac{1}{2}x$, x^2, remainder 4

$\frac{x^2+4}{2x} = \frac{1}{2}x + \frac{4}{2x}$

polynomial, proper rational function

A rational function is proper if $\deg(f) < \deg(g)$, and improper if $\deg(f) \geq \deg(g)$.

If the integrand is an improper function, we can first apply the long division to express it as the sum of a polynomial and a proper rational function.

To integrate a proper rational function, we express it as the sum of partial fractions (if possible).

General Description of the Method

$x^2+2x+4 = (x+2)^2$ ← linear factor $(x+2)(x+2)$

Success in writing a rational function $\frac{f(x)}{g(x)}$ as a sum of partial fractions depends on three things:

- The degree of $f(x)$ must be less than the degree of $g(x)$. That is, the function must be proper. If isn't, divide $f(x)$ by $g(x)$ and work with the remainder term.

must be proper : Improper ⇒ long division. → polynomial + proper

$\frac{5x-3}{x^2-2x-3}$ ← $g(x) = (x+1)(x-3)$

- We must know the factors of $g(x)$. In theory, any polynomial with real coefficients can be written as a product of real linear factors and real quadratic factors. In practice, the factors may be hard to find.
- The values of the undetermined coefficients form a system of n linear equations in n unknowns.

Write $\frac{5x-3}{x^2-2x-3} = \frac{A}{x+1} + \frac{B}{x-3}$? ?

We say a quadratic polynomial is irreducible if it is not a product of two real linear factors. That is, it has no real root.

Eg: $x^2-4 = (x+2)(x-2)$ → reducible

x^2+4 ← irreducible

linear → $(x-1)(x^2+4)$ ← quadratic.

Partial Fraction Decompositions

1. Let $x - r$ be a linear factor of $g(x)$. Suppose that $(x - r)^m$ is the highest power of $x - r$ that divides $g(x)$. Then to this factor, assign the sum of the m partial fractions:

$$\frac{A_1}{(x-r)} + \frac{A_2}{(x-r)^2} + \cdots + \frac{A_m}{(x-r)^m}.$$

   Do this for each distinct linear factor of $g(x)$.

2. Let $x^2 + px + q$ be an irreducible quadratic factor of $g(x)$. In this case, $x^2 + px + q$ has no real roots. Suppose that $(x^2 + px + q)^n$ is the highest power of this factor that divides $g(x)$. Then, to this factor, assign the sum of the n partial fractions:

$$\frac{B_1 x + C_1}{(x^2 + px + q)} + \frac{B_2 x + C_2}{(x^2 + px + q)^2} + \cdots + \frac{B_n x + C_n}{(x^2 + px + q)^n}.$$

   Do this for each distinct quadratic factor of $g(x)$.

Partial Fraction Decompositions

3. Set the original fraction $f(x)/g(x)$ equal to the sum of all these partial fractions. Clear the resulting equation of fractions.
4. Find the values of the undetermined coefficients.

Example: Find the partial fraction decomposition of $\frac{x^2-3x+8}{x^3-4x^2+4x}$. ($f(x)$ / $g(x)$)

$$\frac{x^2-3x+8}{x^3-4x^2+4x} = \frac{x^2-3x+8}{x(x^2-4x+4)} = \frac{x^2-3x+8}{x(x-2)^2}$$

Let $\frac{x^2-3x+8}{x^3-4x^2+4x} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$

$$\frac{x^2-3x+8}{x(x-2)^2} = \frac{A(x-2)^2 + Bx(x-2) + Cx}{x(x-2)^2}$$

So, $x^2 - 3x + 8 = A(x-2)^2 + Bx(x-2) + Cx$

↓ Expand and rearrange in terms of powers of x.

$$= \underline{A+B}\,x^2 + \underline{C-2B-4A}\,x + \underline{4A}$$

Comparing the coefficients.

$$\left.\begin{array}{ll} x^2: & A+B = 1 \\ x^1: & C-2B-4A = -3 \\ x^0: & 4A = 8 \end{array}\right\} \Rightarrow \begin{array}{l} A = 2 \\ B = -1 \\ C = 3 \end{array}$$

So, $\dfrac{x^2-3x+8}{x(x-2)^2} = \dfrac{2}{x} + \dfrac{-1}{(x-2)} + \dfrac{3}{(x-2)^2}$

$$\int \frac{x^2-3x+8}{x^3-4x^2+4x}\,dx = \int \frac{2}{x} - \frac{1}{x-2} + \frac{3}{(x-2)^2}\,dx$$

$\frac{2}{x}$ → $2\ln|x|$

$-\frac{1}{x-2}$ → $-\ln|x-2|$

$\frac{3}{(x-2)^2}$ → $-3(x-2)^{-1}$

$$= 2\ln|x| - \ln|x-2| - \frac{3}{x-2} + C$$

Partial Fraction Decompositions

Find the partial fraction decomposition of $\frac{2x+1}{(x^2+x+1)^2}$.

↑ no real root, irreducible.

Let $\frac{2x+1}{(x^2+x+1)^2} = \frac{Ax+B}{x^2+x+1} + \frac{Cx+D}{(x^2+x+1)^2}$

Multiply both sides by $(x^2+x+1)^2$ ← denominator

$$2x+1 = (Ax+B)(x^2+x+1) + Cx+D$$

$$= Ax^3 + (A+B)x^2 + (A+B+C)x + (B+D)$$

Comparing the coefficients:

$$\left.\begin{aligned} A &= 0 \\ A+B &= 0 \\ A+B+C &= 2 \\ B+D &= 1 \end{aligned}\right\} \Rightarrow \begin{aligned} A &= 0 \\ B &= 0 \\ C &= 2 \\ D &= 1 \end{aligned}$$

$$\frac{2x+1}{(x^2+x+1)} = \frac{2x+1}{x^2+x+1}$$

This means it is already in the partial fraction forms.

Integrate by substitution

$u = x^2+x+1$

$$\frac{2x+1}{(x+1)(x^2+x+1)^2} = \frac{A}{x+1} + \frac{Bx+C}{x^2+x+1} + \frac{Dx+E}{(x^2+x+1)^2}$$

Multiply both sides by $(x+1)(x^2+x+1)^2$,

$$2x+1 = A(x^2+x+1)^2 + (Bx+C)(x+1)(x^2+x+1) + Dx+E(x+1)$$

$$= (A+B)x^4 + (2A+2B+C)x^3 + (3A+2B+2C+D)x^2 + (2A+B+2C+D+E)x + (A+C+E)$$

Comparing the coefficient

$A = -1$

$B = 1$

$C = 0$

$D = 1$

$E = 2$

↓ Exercise.

$$\frac{2x+1}{(x+1)(x^2+x+1)^2} = \frac{-1}{x+1} + \frac{x}{x^2+x+1} + \frac{x+2}{(x^2+x+1)^2}$$

Ex: Integrate it.

↓ If $u = x^2+x+1$

$du = 2x+1\,dx$

$$\frac{x}{x^2+x+1} = \frac{1}{2}\left(\frac{2x+1-1}{x^2+x+1}\right)$$

$$= \frac{1}{2}\left(\frac{2x+1}{x^2+x+1}\right) - \frac{1}{2}\left(\frac{1}{x^2+x+1}\right)$$

substitution

completing the square

Lec 2.

Examples

$$\int \frac{x^2 + 4x + 1}{(x-1)(x+1)(x+3)}\,dx$$

deg 2 ← x^2 ; deg 3 ← $(x-1)(x+1)(x+3)$; proper

Let $\dfrac{x^2+4x+1}{(x-1)(x+1)(x+3)} = \dfrac{A}{x-1} + \dfrac{B}{x+1} + \dfrac{C}{x+3}$

Multiply both sides by $(x-1)(x+1)(x+3)$, we get

$$x^2+4x+1 = A(x+1)(x+3) + B(x-1)(x+3) + C(x-1)(x+1)$$

$$= (A+B+C)x^2 + (4A+2B)x + (3A-3B-C)$$

Comparing the coefficient,

$$\left.\begin{aligned} A+B+C &= 1 \\ 4A+2B &= 4 \\ 3A-3B-C &= 1 \end{aligned}\right\} \Rightarrow A=\frac{3}{4},\ B=\frac{1}{2},\ C=\frac{-1}{4}$$

$$\text{So } \int \frac{x^2+4x+1}{(x-1)(x+1)(x+3)}\,dx = \int \frac{3}{4}\left(\frac{1}{x-1}\right)+\frac{1}{2}\left(\frac{1}{x+1}\right)-\frac{1}{4}\left(\frac{1}{x+3}\right)dx$$

$$= \frac{3}{4}\ln|x-1|+\frac{1}{2}\ln|x+1|-\frac{1}{4}\ln|x+3| + D$$

↑ to avoid the confusion with C above.

Examples

$$\int \frac{6x+7}{(x+2)^2}\,dx$$

deg 1 < deg 2 — proper

Let $\frac{6x+7}{(x+2)^2} = \frac{A}{x+2} + \frac{B}{(x+2)^2}$

Multiplying both sides by $(x+2)^2$,

$6x+7 = A(x+2) + B$

$= Ax + (2A+B)$

Comparing the coefficients,

$A = 6$

$2A+B = 7 \Rightarrow B = -5$

$$5. \int \frac{6x+7}{(x+2)^2}\,dx = \int 6\left(\frac{1}{x+2}\right) - 5\left(\frac{1}{(x+2)^2}\right)dx$$

$$= 6\ln|x+2| - 5\left(\frac{(x+2)^{-1}}{-1}\right) + C$$

$$= 6\ln|x+2| + \frac{5}{x+2} + C$$

Examples

$$\int \frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3}\,dx$$

deg 3

deg 2

improper

$$\begin{array}{r|l} & 2x \\ \hline x^2-2x-3 & 2x^3-4x^2-x-3 \\ & (-)\ 2x^3-4x^2-6x \\ \hline & 5x-3 \end{array}$$

quotient

remainder.

$$\frac{2x^3-4x^2-x-3}{x^2-2x-3} = \underbrace{2x}_{\text{polynomial}} + \underbrace{\frac{5x-3}{x^2-2x-3}}_{\text{proper rational function.}}$$

(partial fraction decomposition)

$$\frac{5x-3}{x^2-2x-3} = \frac{5x-3}{(x-3)(x+1)}$$

Let $\frac{5x-3}{x^2-2x-3} = \frac{A}{x-3} + \frac{B}{x+1}$

$$\Rightarrow 5x-3 = A(x+1) + B(x-3)$$

$$= (A+B)x + (A-3B)$$

Comparing the coefficients,

$$\left.\begin{aligned} A+B &= 5 \\ A-3B &= -3 \end{aligned}\right\} \Rightarrow \begin{aligned} A &= 3 \\ B &= 2 \end{aligned}$$

So $\int \frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3}\, dx = \int 2x + 3\left(\frac{1}{x-3}\right) + 2\left(\frac{1}{x+1}\right) dx$

$= x^2 + 3\ln|x-3| + 2\ln|x+1| + C$

Examples

$$\int \frac{-2x+4}{(x^2+1)(x-1)^2}\,dx$$

< deg 1

proper

irreducible quadratic factor

< deg 4

linear factor with power of 2

Let $\frac{-2x+4}{(x^2+1)(x-1)^2} = \frac{Ax+B}{(x^2+1)} + \frac{C}{x-1} + \frac{D}{(x-1)^2}$

$$\Rightarrow -2x+4 = (Ax+B)(x-1)^2 + C(x^2+1)(x-1) + D(x^2+1)$$

$$= (A+C)x^3 + (-2A+B-C+D)x^2$$

$$+ (A-2B+C)x + (B-C+D)$$

Comparing the coefficients,

$$\left.\begin{aligned} A+C &= 0 \\ -2A+B-C+D &= 0 \\ A-2B+C &= -2 \\ B-C+D &= 4 \end{aligned}\right\} \Rightarrow \begin{aligned} A &= 2, \quad B = 1 \\ C &= -2, \quad D = 1 \end{aligned}$$

So $$\int \frac{-2x+4}{(x^2+1)(x-1)^2}\,dx = \int \frac{2x+1}{x^2+1} - 2\left(\frac{1}{x-1}\right) + \frac{1}{(x-1)^2}\,dx$$

substitute $u = x^2+1$ for the first term; rewrite $\frac{1}{x-1}$ as $(x-1)^{-1}$

$$= \int \frac{2x}{x^2+1} + \frac{1}{x^2+1} - 2\left(\frac{1}{x-1}\right) + (x-1)^{-2}\,dx$$

$$= \ln|x^2+1| + \arctan x$$
$$-2\ln|x-1| - \frac{1}{(x-1)} + K$$

Examples

$$\int \frac{1}{x(x^2+1)^2}\,dx$$

deg 0 ← 1 ; deg 5 ← $x(x^2+1)^2$; proper

x: linear factor

x^2+1: irreducible quadratic factor of power 2.

Let $\dfrac{1}{x(x^2+1)^2} = \dfrac{A}{x} + \dfrac{Bx+C}{x^2+1} + \dfrac{Dx+E}{(x^2+1)^2}$

$0x^4+0x^3+0x^2+0x+1$

$$\Rightarrow \quad 1 = A(x^2+1)^2 + (Bx+C)\,x(x^2+1) + (Dx+E)\,x$$

$$= (A+B)x^4 + Cx^3 + (2A+B+D)x^2 + (C+E)x + A$$

Comparing the coefficients,

$$\left.\begin{aligned} A+B &= 0 \\ C &= 0 \\ 2A+B+D &= 0 \\ C+E &= 0 \\ A &= 1 \end{aligned}\right\} \Rightarrow \begin{aligned} A &= 1 \\ B &= -1 \\ C &= 0 \\ D &= -1 \\ E &= 0 \end{aligned}$$

$$\int \frac{1}{x(x^2+1)^2}\,dx = \int \frac{1}{x} - \frac{x}{x^2+1} - \frac{x}{(x^2+1)^2}\,dx$$

Substitution $u = x^2+1$

$$= \ln|x| - \frac{1}{2}\ln|x^2+1| + \frac{1}{2}\left(\frac{1}{x^2+1}\right) + k$$

Examples

$$\int \frac{4x^2 - 15x + 29}{(x-5)(x^2 - 4x + 13)}\, dx$$

Exercise.

linear

$b^2 - 4ac$

$= (-4)^2 - 4(1)(13)$

$= -ve$ (no real roots)

irreducible.

Let $\dfrac{4x^2 - 15x + 29}{(x-5)(x^2 - 4x + 13)} = \dfrac{A}{x-5} + \dfrac{Bx + C}{x^2 - 4x + 13}$

Ans: $3 \ln|x-5| + \frac{1}{2} \ln|x^2 - 4x + 13| + \frac{4}{3} \arctan\left(\frac{x-2}{3}\right) + K$

Other Ways to Find the Coefficients

Using differentiation: Find A, B, and C in the equation

$$\frac{x-1}{(x+1)^3} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3}.$$

$$\Rightarrow \quad x-1 = A(x+1)^2 + B(x+1) + C$$

Let $x=-1$, $-1-1 = A(-1+1)^2 + B(-1+1) + C$

$$C = -2$$

$$x-1 = A(x+1)^2 + B(x+1) - 2$$

differentiate w.r.t. x

$$1 = 2A(x+1)(1) + B$$

Let $x=-1$, $1 = 2A(-1+1) + B \Rightarrow B = 1$

Differentiate w.r.t. x again,

$$0 = 2A \Rightarrow A = 0$$

So, $$\frac{x-1}{(x+1)^3} = \frac{1}{(x+1)^2} - \frac{2}{(x+1)^3}$$

Other Ways to Find the Coefficients

Assigning numerical value to x: Find A, B, and C in the equation

When all the factors are linear.

$$\frac{x^2+1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}.$$

$= \frac{1}{x-1} - \frac{5}{x-2} + \frac{5}{x-3}$

$\Rightarrow x^2+1 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$

Let $x=1$, $1^2+1 = A(1-2)(1-3) + B(1-1)(1-3) + C(1-1)(1-2)$

$\therefore A = 1$

Let $x=2$, $2^2+1 = B(2-1)(2-3) \Rightarrow B = -5$

Let $x=3$, $3^2+1 = C(3-1)(3-2) \Rightarrow C = 5$